Intermittent rate coding without short-term plasticity

Abstract

Working memory is thought to rely either on persistent spiking activity, sustained by slow NMDA receptor currents, or on activity-silent synaptic traces. Recently, population recordings from non-human primate prefrontal cortex showed that the remembered item can be decoded only intermittently on single trials, with Off periods in which decoding falls to chance. These Off periods have been taken as evidence that memories are carried by short-term synaptic plasticity when spiking activity fails to hold them. Here, we show that intermittent decoding is also expected from a canonical attractor network without plastic synapses. When a shared input (such as fluctuating attention) briefly drops, the network falls silent and decoding fails. However, NMDA receptor currents, which decay over 150 ms as measured in cortical neurons, bridge the silence and the memory is reinstated when the attentional input returns. Fitted to population recordings from non-human primate prefrontal cortex, this model reproduces quantitatively the number and duration of On and Off states while retaining the memory. In the recordings, we found that Off states are more frequent when fewer or less selective neurons are decoded, so their frequency partly reflects how well the recorded sample reads out the memory. Without refitting, the model predicts this dependence for decoded populations it was never fitted to. Intermittent decoding is therefore compatible with persistent activity and does not, on its own, require activity-silent mechanisms. Finally, unlike a model in which short-term facilitation holds the memory, the NMDA model also predicts that errors increase with the time spent Off, as we found in the recordings. All together, these results favour persistent activity bridged by a short-lived NMDA trace over activity-silent storage in short-term facilitation.

Introduction

Two main mechanisms have been proposed to hold information in working memory. In persistent-activity models, recurrent excitation keeps a cue-selective population firing through the delay, and slow NMDA receptor currents make this activity stable (Amit and Brunel, 1997; Wang, 1999, 2001; Compte et al., 2000). In activity-silent models, the memory is instead stored in short-term synaptic facilitation and needs no ongoing spikes for memory maintenance (Mongillo et al., 2008; Stokes, 2015).

Large-scale recordings now make it possible to test these accounts on single trials. Recent studies agree that working-memory information in prefrontal cortex fluctuates within a trial, but disagree on what the fluctuations mean. Panichello et al. (2024) found that decoding alternates between On states, when the cue can be read out, and Off states, when it cannot, and concluded that synaptic changes carry the memory through Off states. Tao and Libedinsky (2024) found that information carried by spiking activity is reduced but still present between spike bursts. They also found rare periods of 43–414 ms in which all simultaneously recorded selective neurons were silent for longer than expected by chance, and concluded that short-term plasticity is needed only in a very small fraction of trials. Mozumder et al. (2026) found that persistent activity is asynchronous across neurons, with no population silences beyond chance. In their recordings, decoding Off states came with reduced but not abolished tuning and were synchronized between prefrontal and parietal cortex, and the authors interpreted them as fluctuations in the fidelity of a persistent code.

Here, we ask whether a single persistent-activity mechanism can account for these observations and propose that intermittent decoding does not require short-term plasticity and is in fact consistent with the NMDA account. In our model, the memory is held in persistent activity supported by recurrent attractors (Wang, 2002; Wong and Wang, 2006). However, fluctuations in a common input, which we interpret as a fluctuating attentional signal, briefly drops and silences the network. In between silences, slow NMDA currents carry the memory across, so the cued population resumes firing when the attentional signal returns. We read out the model with the same decoding and On/Off criteria as Panichello et al. (2024) and fit how often, for how long and how deeply attention lapses to the On/Off statistics of these recordings. The model reproduces quantitatively the number and length of On and Off states without memory loss. Reanalyses of the recordings showed that Off states are more frequent when fewer (or less selective) neurons are decoded, so their frequency partly reflects how well the recorded sample reads it out. Without refitting, the attractor model predicts this dependence for decoded populations it was never fitted to. Intermittent decoding is therefore compatible with persistent activity sustained by NMDA currents, without short-term plasticity. Finally, unlike a model in which short-term facilitation carries the memory, the NMDA model predicts that errors grow with the time spent Off, as we found in the recordings

Results

Input dips produce Off states and NMDA carries the memory across them

The model is a classic two-pool attractor network (Wang, 2002; Wong and Wang, 2006), with two memory pools, cued and uncued, that excite themselves through NMDA currents and compete through shared inhibition (Methods). After the cue, the cued pool stays active and decoding stays On for the whole delay (Fig. 1a, first second after the cue). The key addition to this classic model is a common input to both pools that fluctuates, which we interpret as an attentional signal, since attention is known to wax and wane on sub-second time scales (Fiebelkorn and Kastner, 2019, see Discussion). With brief dips in this input, both pools fall silent and decoding drops to chance (Fig. 1a, after 1,000 ms). During each dip, the NMDA gating of the cued pool decays with a time constant of 150 ms and stays higher than that of the uncued pool. When the input returns, lagging feed-forward inhibition gives both pools a brief kick, and the pool with more NMDA gating wins the competition. Off states therefore do not erase the memory and NMDA gating carries it across each silence, ensuring a correct memory of the cue at the end of the delay.

To compare the model with the data, we turned its activity into a decoding z score: the difference between the cued and uncued pool rates in 100 ms windows, multiplied by a gain (c) and added to noise (η) whose variance and temporal correlations match those of the decoder output in the data (Methods). Importantly, the gain was not fitted to the On/Off statistics. Instead, it was calibrated on a single quantity measured in the data, the typical z value when decoding is high, which we call the decoding strength (Methods). On and Off states (red and cyan in Fig. 1a) are then labelled with the same thresholds as in Panichello et al. (2024) (red and cyan dashed lines in Fig. 1a). Read out this way, the model alternates between On and Off states (Fig. 1a, bottom). Both length distributions peak near 100 ms and have long tails (Fig. 1b), qualitatively like those in the recordings (Panichello et al., 2024). In the next sections, we ask whether the model also matches the data quantitatively, including for decoders built from different sets of neurons.

On/Off statistics depend on which neurons are decoded

So far we considered decoders that use all recorded neurons. We found, however, that On/Off statistics depend strongly on which neurons the decoder uses, which we call the decoded population

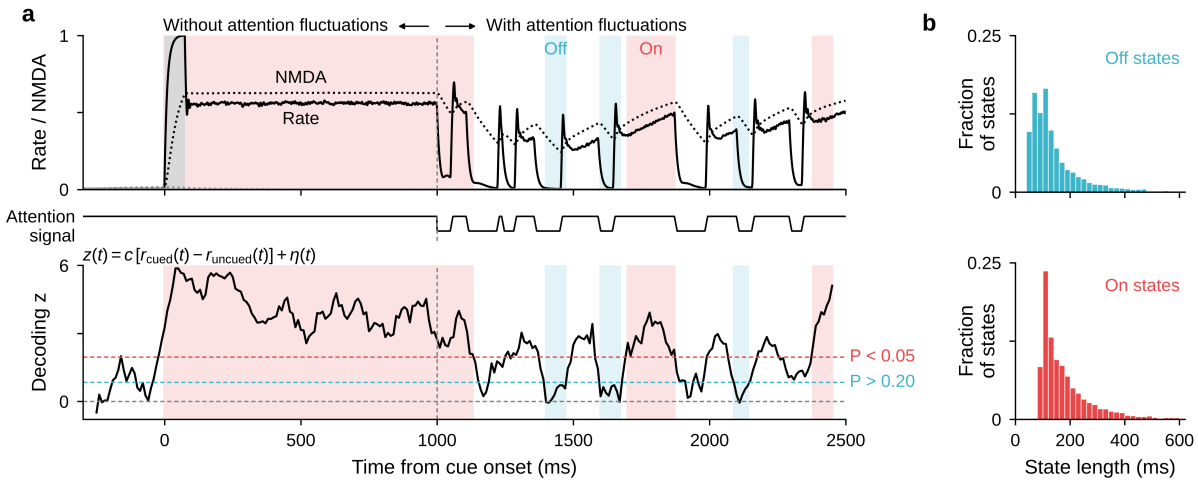


Figure 1: Input dips create On and Off states, and NMDA carries the memory. a, Example trial. Top: rates of the cued (black) and uncued (grey) pools, relative to the maximum rate, and their NMDA gating (dotted); grey shading marks the cue. Middle: common input (attention signal), constant for the first 1,000 ms after the cue and with dips afterwards. The dip-free first second is shown for illustration only; in all other simulated trials, dips occur throughout the delay. Bottom: decoding z score, with On (red) and Off (cyan) states. Dashed lines: On threshold ($z = 1.96$) and Off threshold ($z = 0.84$), as in Panichello et al. (2024). **b**, Length distributions of Off (top) and On (bottom) states in the model.

(Fig. 2, black circles). Decoders using the most cue-selective neurons spend much more of the delay On, and less Off, than decoders using the same numbers of randomly chosen neurons, which also show more Off states per trial. The decoded populations differ mainly in decoding strength, which is much higher for the most selective neurons. How often a memory appears Off therefore depends partly on how well the recorded neurons read it out. This matters for interpreting Off states as evidence for activity-silent memory: a decoder trained on a limited sample of neurons can fall to chance while the memory is still present in the activity of other, potentially unrecorded neurons. The measured fraction of Off states is therefore an upper bound on how often the memory leaves spiking activity. We then asked if our model could explain this dependence.

A model fitted to all neurons predicts this dependence

If Off states come from shared dips in network activity, the same dynamics, read out with a higher or lower decoding strength, should reproduce the change across decoded populations. To test this, we fitted the dynamics only to the decoder using all neurons in 25 sessions (Methods). The model matches the data well in Off states per trial (3.1 vs 3.3), Off median length (110 vs 112 ms), On states per trial (1.9 vs 2.1) and On median length (140 vs 134 ms) (Fig. 2, grey band). We then kept the dynamics fixed and, for each of the six other decoded populations, changed only the read-out without refitting any parameters. Its gain was calibrated on that population's decoding strength and its noise on that population's pre-cue z (Methods). Without any refitting, the model predicts quantitatively the shift between selective and random neurons.

Errors increase with time Off, as expected from an NMDA trace

So far, we analysed correct trials only, but error trials can help distinguish between the two accounts of working memory. NMDA currents and short-term synaptic facilitation can both carry a memory across brief Off states, but on different time scales: NMDA gating decays within ~150 ms, whereas facilitation decays over ~1.5 s (Mongillo et al., 2008; Mi et al., 2017), comparable to the whole memory period (the delay ended 1.4–1.6 s after cue onset). A short

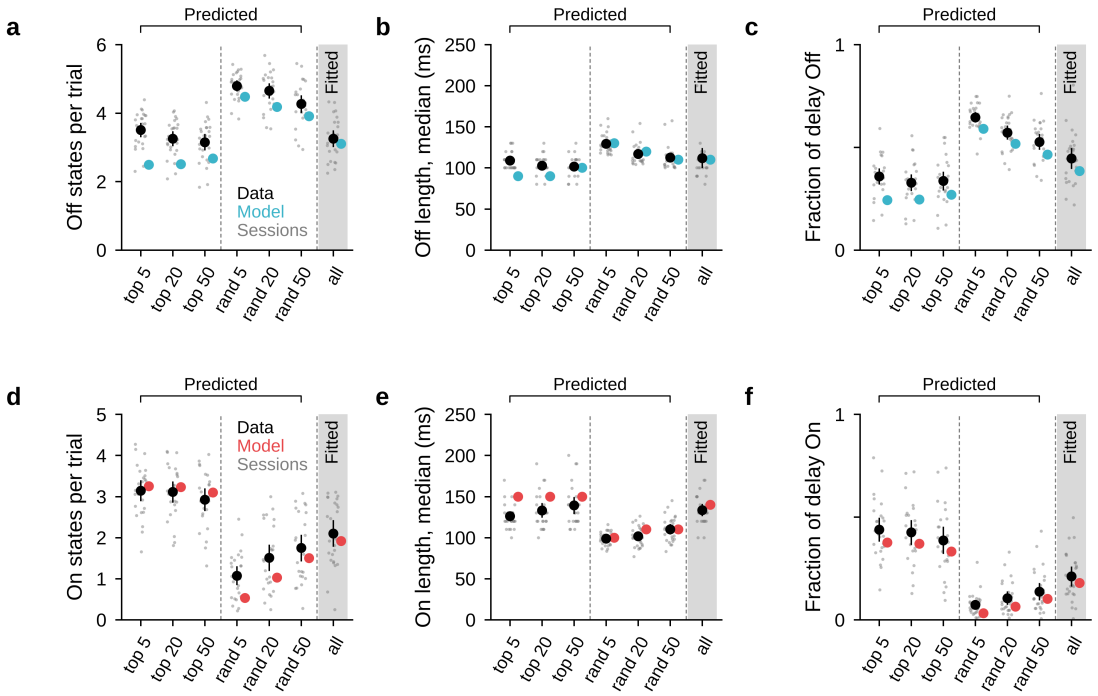


Figure 2: The model predicts how On/Off statistics change with the decoded population. Decoders using the most cue-selective neurons (top 5, 20, 50) are On more often and Off less often than decoders using the same numbers of random neurons (rand 5, 20, 50). Black circles: data, mean ± 95% CI (2 SEM) over 25 sessions. Grey dots: single sessions. Cyan and red circles: model, Off and On statistics. The model was fitted to all neurons only (grey bar); the other six decoded populations are predictions obtained by changing only the read-out gain and noise.

trace predicts that the memory is occasionally lost when an Off period outlasts it, so that errors become more frequent the more time a trial spends Off. A trace that outlasts the delay predicts no such relation. Indeed, when we replaced NMDA gating by facilitation in our model and refitted it, it reproduced the On/Off statistics about as well as the NMDA model but lost the memory in only 9 of 100,000 simulated trials, whatever the Off time (STP model; Methods). In a test with a single dip of the common input per trial, the NMDA model lost the memory whenever the dip lasted longer than about 375 ms, whereas the STP model held it for dips of up to 1 s. Read out as the data, memory in the NMDA model, but not in the STP model, therefore declined steeply with Off duration (Fig. 3a).

We tested this prediction by comparing On/Off statistics between correct and error trials (Methods). Within sessions, the odds of an error increased with the total Off time in the delay and with the mean length of Off states, and decreased with the total On time, but did not depend on the number of Off states (Fig. 3b). Errors thus depended on how long the memory was Off, not on how often it went Off: frequent Off states are harmless as long as each is brief. The effect was clear in monkeys H and J (odds ratios per SD of total Off time 1.61, $p = 1 \times 10^{-4}$, and 1.68, $p = 0.005$) but weak in monkey A (1.11, $p = 0.13$).

Finally, we asked which time constant of the trace best explains how performance falls with Off time. We varied τ_N in the fitted NMDA model, leaving all other parameters unchanged, and simulated each version with its own fitted dips, read out as the data. In these simulations, the memory is lost when a single Off period outlasts the NMDA trace and the cued pool cannot restart (with $\tau_N = 150$ ms, single dips longer than about 375 ms; Fig. 3a), whereas long Off times made of several brief Off states are mostly bridged. We linked the probability that the memory was held to the probability of a correct response, assuming that the monkeys guess among the 8 locations after a memory loss and that other errors do not depend on Off time (Methods).

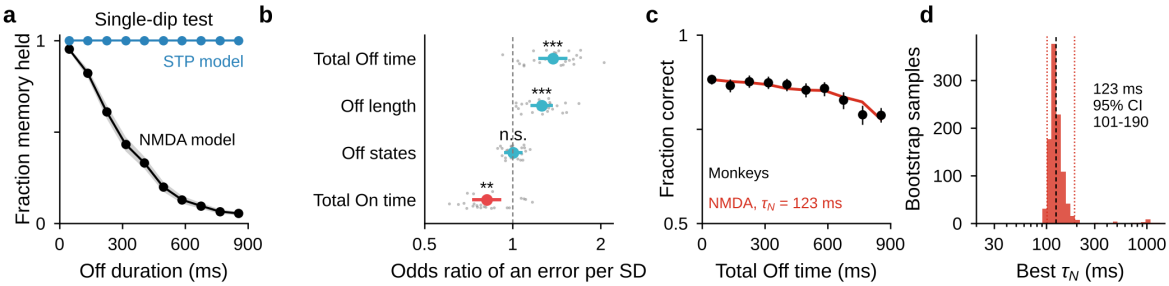


Figure 3: Errors increase with time Off, and the decline in performance points to a trace of about 120 ms. a, Single-dip test, read out as the data: fraction of trials with the memory held against the Off duration in the delay, for the NMDA model (black) and the STP model (blue), with one dip of the common input of 0 to 1 s per trial (12,300 trials per model; Methods). Shading: 95% Wilson interval. **b**, Odds ratio of an error per within-session SD of four statistics of the delay (500–1,400 ms after cue onset): total Off time, mean Off-state length, number of Off states and total On time. Coloured circles and lines: fixed effect and 95% CI of a logistic mixed model, fitted for each statistic to all 25 sessions (Methods). Grey dots: single sessions (fixed effect plus the session's random slope). Likelihood-ratio tests: total Off time, $p = 1 \times 10^{-5}$; Off length, $p = 7 \times 10^{-5}$; number of Off states, $p = 0.89$; total On time, $p = 0.001$. $^{***}p < 0.001$, $^{**}p < 0.01$, n.s. not significant. **c**, Fraction correct against total Off time in the delay, all 15,468 trials pooled, in 90 ms bins (black; 95% Wilson interval), and the NMDA model with the best-fitting τ_N of 123 ms (red), linked to behaviour by P(correct) = $p_0 \times$ [P(held) + P(lost)/8] (Methods). **d**, Best τ_N in 1,000 bootstraps over sessions. Dashed: fit to all data. Dotted: 95% interval (101–190 ms).

With the best-fitting τ_N of 123 ms, the model reproduced the decline of the pooled fraction correct (Fig. 3c). A bootstrap over sessions gave a 95% interval of 101–190 ms (Fig. 3d), and the estimate changed little with the assumed guessing rate (Methods). The decline of performance with Off time thus calls for a trace of about 120 ms, close to measured NMDA receptor kinetics and to the 150 ms assumed in the fitted model, and far shorter than the ∼1.5 s of facilitation.

Discussion

Recent large-scale recordings agree that prefrontal decoding of a remembered item is intermittent on single trials, but differ on whether Off states mean that the memory leaves spiking activity and is held in short-term synaptic plasticity (Panichello et al., 2024; Tao and Libedinsky, 2024; Mozumder et al., 2026). We showed that the same On/Off statistics arise in a conventional attractor network with fixed synapses. In the model, Off states occur when a common input briefly drops, and the memory survives each silence because NMDA receptor currents decay slowly enough to bias which pool restarts. Fitted to the decoder trained on the empirical data, the model reproduced the number and length of On and Off states and kept the memory in 98% of trials. Unlike a model in which short-term facilitation carries the memory, it also predicted that errors grow with the time spent Off, and they did in the recordings.

In the recordings, Off states were more frequent and On states rarer when the decoder used random rather than selective neurons. This showed that a decoder falls to chance whenever the memory signal in its sample is weak relative to its noise, not only when the memory is absent from the network. The fraction of time Off measured from a limited sample is therefore an upper bound on how often the memory leaves spiking activity, and comparisons of Off-state statistics across studies or brain areas need to take the decoded population into account. The model captured this shift without refitting by changing only the read-out gain and noise, both calibrated from the data.

Our account is consistent with the other recent recordings of Off states. Mozumder et al. (2026) found that Off states were synchronized between prefrontal and posterior parietal cortex,

came with lower firing rates, and showed no obvious rhythmicity. A common input that reaches several areas and fluctuates irregularly, as in our fit, would produce such coordinated, aperiodic Off states. They also found that tuning was reduced by about 78% but not abolished during Off states. Tao and Libedinsky (2024) likewise found that information is reduced but present between gamma bursts. Our model is compatible with both: NMDA currents carry the memory across full silences, but the model does not need them, since the memory is held by persistent activity without any Off state while attention is sustained (Fig. 1a, first second after the cue).

We interpreted the common input as an attentional signal that is high when attention is deployed on the memorandum and drops during lapses. This links intermittent decoding to evidence that sustained attention samples rhythmically rather than continuously. In humans, detection of threshold stimuli depends on the phase of ongoing EEG oscillations near 7 Hz (Busch et al., 2009), especially at attended locations (Busch and VanRullen, 2010). When attention is divided between two locations, detection at each fluctuates at about 4 Hz, in antiphase between locations (Landau and Fries, 2012). In monkeys, frontal eye field and lateral intraparietal activity alternates at theta frequencies (3–8 Hz) between periods of enhanced perception and periods in which attention is more likely to shift (Fiebelkorn et al., 2018), and human intracranial recordings show similar theta-rhythmic sampling in frontoparietal cortex (Helfrich et al., 2018). These findings have been unified in a rhythmic theory of attention, in which theta rhythms alternate between sampling and shifting (Fiebelkorn and Kastner, 2019). Because attention and working memory share prefrontal mechanisms (Awh and Jonides, 2001; Panichello and Buschman, 2021), lapses of attention could plausibly reach the memory network. In the fitted model, the attention signal recovers about five times per second, or one cycle every ∼200 ms. This is in the low theta range reported for attentional sampling (Landau and Fries, 2012; Fiebelkorn et al., 2018; Helfrich et al., 2018) and only slightly slower than the ∼7 Hz phase effects on detection (Busch et al., 2009; Busch and VanRullen, 2010). However, the fitted signal was not periodic. A direct test would be to ask whether Off-state onsets lock to the phase of theta activity in the prefrontal local field potential, or whether probes presented during Off states are detected less often.

In our model, the memory is carried across each silence by NMDA gating that decays within ∼150 ms and reflects recent firing rather than plasticity. This time constant lies within the measured decay times of NMDA receptor currents, which are slowest for receptors containing the GluN2B (NR2B) subunit (Vicini et al., 1998), and GluN2B-containing NMDA receptors are required for persistent firing in primate prefrontal cortex (Wang et al., 2013). Off states last about 100 ms, within the time an NMDA trace can bridge, so no longer-lasting mechanism is needed to explain them. The behavioural data point the same way. A memory held by facilitation decaying over ∼1.5 s should hardly be affected by a few hundred more milliseconds Off, as the STP model showed. We found that errors increased with the time spent Off and with the length of Off states, but not with their number (Fig. 3). This is expected from a trace that bridges brief silences but not long ones. Fitted to the monkeys' performance, the time constant of this trace was about 120 ms (95% CI 101–190 ms), close to measured NMDA receptor kinetics and far shorter than the ∼1.5 s of facilitation. Activity-silent mechanisms may still matter on longer time scales, for example to carry information between trials, when prefrontal memory traces are reactivated before the next stimulus (Barbosa et al., 2020). Notably, this between-trial trace is not a voluntary memory. It does not require attention and shows up as involuntary serial biases in behaviour, rather than as the information the subject actively maintains. Importantly, the model restarts the memory with a brief transient: when the common input returns, excitation precedes lagging feed-forward inhibition by ∼15 ms. Without such a transient, pools silenced by a dip often failed to restart. We implemented it as feed-forward inhibition, a common circuit motif, but recovery of input synapses from depression during a dip would produce a similar onset overshoot. Persistent activity and slower silent traces are therefore not mutually exclusive.

In sum, intermittent decoding of working memory is expected from an attractor network with slow NMDA currents and a fluctuating, attention-like input, and its statistics depend on how

well the recorded neurons read out the memory, while errors increase with the time spent Off, as expected from the limited duration of the NMDA trace. These results reconcile intermittent rate coding with persistent-activity accounts and link Off states to the rhythmic but irregular nature of attentional sampling.

Methods

Rate model

The network is the reduced two-pool attractor model of Wong and Wang (2006), with the changes described below. Two excitatory memory pools, $i \in \{\text{cued}, \text{uncued}\}$, have rates r_i and NMDA gating s_i. They share an inhibitory population r_I and receive a feed-forward inhibition q that follows the common input $m(t)$ with a lag. Parameters in bold were free during fitting; all others were fixed.

$$\tau_r \frac{dr_i}{dt} = -r_i + \varphi(x_i), \tag{1}$$

$$x_i = \boldsymbol{w_s} s_i + w_a r_i - w_I r_I + I_0\, m(t) - w_{ff}\, q + \text{cue}_i + \boldsymbol{\sigma}\xi_i, \tag{2}$$

$$\tau_N \frac{ds_i}{dt} = -s_i + \beta\, r_i(1 - s_i), \tag{3}$$

$$\tau_I \frac{dr_I}{dt} = -r_I + w_{EI}\,(r_{\text{cued}} + r_{\text{uncued}})\,, \tag{4}$$

$$\tau_{ff} \frac{dq}{dt} = -q + m(t). \tag{5}$$

The transfer function φ is a logistic sigmoid, $\varphi(x) = r_{\max}/\big(1 + \exp(-(x-1)/\kappa)\big)$, with maximum rate $r_{\max}$, threshold 1 and slope κ, and ξ_i are independent Ornstein–Uhlenbeck noises ($\tau_\xi = 10$ ms). Fixed parameters: $r_{\max} = 150$, $\kappa = 0.06$, $w_a = 0.0141$, $\beta = 0.02$, $w_I = 0.04$, $w_{EI} = 0.5$, $w_{ff} = 0.8$, $I_0 = 1.496$; $\tau_r = 10$ ms, $\tau_I = 1$ ms, $\tau_{ff} = 15$ ms, $\tau_N = 150$ ms. The implementation differs from Wong and Wang (2006) in three respects: the transfer function is a saturating sigmoid, competition between the pools goes through an explicit inhibitory population, and a lagging feed-forward inhibition is added. Parameter values could therefore not be taken directly from their study. We chose them by hand so that each pool is bistable, with a low state near 0 Hz and a high state of about 85 Hz, and so that shared inhibition allows only one pool to be active at a time. The cue is a 50 ms input of amplitude 1 to the cued pool, after 300 ms of baseline, followed by a 1,600 ms delay. Equations were integrated with Euler steps of 0.1 ms. The memory was scored correct when, in the last 100 ms, the NMDA gating of the cued pool exceeded the uncued pool's by more than 0.05. We scored NMDA gating rather than rates because the network may be silent at the end of the delay, but a strong attentional input during the response period can restart the pool with more NMDA gating and recover the correct response with virtually no memory loss (not shown).

Our main modification to this model is the common input $m(t)$ to both pools, which represents fluctuating attention. It is 1 except during dips, where it drops to $1 - \boldsymbol{\delta}$ with 10 ms ramps. Dip lengths are gamma-distributed (mean $\boldsymbol{\mu_d}$, shape $\boldsymbol{k}$), and the gaps between dips are exponential with mean $1/\boldsymbol{\lambda_d}$.

Data

We reanalysed the recordings of Panichello et al. (2024): 25 sessions, correct trials only (error trials were used only in the behavioural analysis), all cue locations (13,190 trials; 329 units per session on average). For each time bin, an L2-regularised logistic decoder ($\lambda = 10{,}000$) was evaluated with approximate leave-one-out and z-scored against 30 shuffled label sets per cue pair. Decoded populations were all neurons, the k most cue-selective neurons (ranked by split-half

ANOVA), and k randomly chosen neurons (10 draws, averaged within sessions), with $k = 5$, 20 and 50.

Behavioural analysis

To relate On/Off states to behaviour, we also labelled error trials (15,468 trials in total, 2,278 errors). Error trials were decoded with decoders fitted on all correct trials of the same cue pair, so they were also out of sample, and their rates were z-scored with the mean and SD of correct trials. For each trial, we computed the total Off time and the total On time in the delay (500–1,400 ms after cue onset), the number of Off states and their mean length (13,962 trials with at least one Off state). For each statistic, the probability of an error was modelled with a logistic mixed model, error $\sim$ z(stat) + t + t^2 + monkey + (1 + z(stat) || session), where z(stat) is the statistic z-scored within each session, so that only within-session variation is used, t is the time within the session, and each session has a random intercept and an uncorrelated random slope. Monkey was a fixed effect, as three monkeys are too few to estimate a random effect. Models were fitted with the Laplace approximation and tested with likelihood-ratio tests against the same model without the statistic. For the fit of τ_N, trials were pooled over sessions and binned by Off time in 90 ms bins (one tenth of the delay window).

STP model

To compare our model with a short-term plasticity account, we replaced the NMDA gating s_i in the recurrent input by a saturating synaptic activation without a time constant of its own, $S_i = g_i/(1 + g_i)$, with $g_i = \beta r_i u_i/U$, where u_i is Tsodyks–Markram short-term facilitation (Mongillo et al., 2008; Mi et al., 2017):

$$\tau_f \frac{du_i}{dt} = U - u_i + \tau_f U(1 - u_i) r_i,$$

with $U = 0.3$ and $\tau_f = 1.5$ s. To simplify fitting and to give the STP model the best chance to fit the data, synaptic depression was switched off: with depression ($\tau_d = 0.3$ s), each re-ignition ended too early and On states were too short (112 vs 172 ms in the data). All other parameters were as in the NMDA model, and the STP model was fitted in the same way to the decoder using all neurons, with an additional penalty for population spikes before the cue. Best fit: $\boldsymbol{w_s} = 0.69$, $\boldsymbol{\lambda_d} = 5.5$/s, $\boldsymbol{\mu_d} = 212$ ms, $\boldsymbol{k} = 0.70$, $\boldsymbol{\delta} = 0.39$, $\boldsymbol{\sigma} = 0.005$. Both models matched the On/Off statistics about equally well (loss 9.7 for the NMDA model and 10.4 for the STP model, on 2,112 trials) and held the memory in nearly all trials.

Simulations of memory against Off time

To compare with the behaviour, each fitted model was then simulated with its own random dips (100,000 trials per model); the memory was held in 98.5% (NMDA model) and 99.99% (STP model) of these trials. In the single-dip test, each trial had one dip of the common input, starting 500 ms after cue onset, with the model's fitted depth and a length from 0 to 1,000 ms in 25 ms steps (300 trials per length, 12,300 trials per model). Trials were read out with the all-units gain and noise of the fitted model, and the memory was scored as in the NMDA model, using the facilitation u_i instead of the NMDA gating in the STP model. Off time was computed as in the data.

NMDA time constant fitted to performance

The fitted NMDA model, with its own dips and the all-units read-out, was simulated with τ_N from 20 to 1,500 ms (17 values, 30,000 trials each) and all other parameters unchanged, so the On/Off statistics drift away from their fitted values as τ_N changes. For each τ_N, the probability

that the memory was held was computed in the ten Off-time bins and interpolated in log τ_N. The model was linked to behaviour by P(correct) = p_0 × [P(held) + G × P(lost)], where G = 1/8 is the chance of a correct guess among the 8 cue locations after the memory is lost, and p_0 is the performance when the memory is held, which absorbs errors that do not depend on Off time. τ_N and p_0 were fitted by maximum likelihood to all 15,468 trials pooled (best p_0 = 0.887). Confidence intervals came from 1,000 bootstraps over sessions (25 sessions drawn with replacement, their trials pooled and refitted). The estimate depended little on G: the best τ_N was 129, 118 and 102 ms for G = 0, 1/4 and 1/2.

Fitting

Six parameters were fitted with the covariance matrix adaptation evolution strategy (CMA-ES; Hansen and Ostermeier, 2001) to the decoder using all neurons: $\boldsymbol{w_s}$, $\boldsymbol{\lambda_d}$, $\boldsymbol{\mu_d}$, $\boldsymbol{k}$, $\boldsymbol{\delta}$ and $\boldsymbol{\sigma}$. The NMDA decay time was not fitted but fixed at τ_N = 150 ms, within the range measured in cortex (Vicini et al., 1998; Wang et al., 2013). The loss combined the squared scaled errors of six statistics (On and Off states per trial, mean lengths and fractions of the delay), the squared Hellinger distance between model and data length distributions (states ≥ 40 ms), and a penalty when the memory was correct in fewer than 98% of trials. Best fit: $\boldsymbol{w_s}$ = 1.30, $\boldsymbol{\lambda_d}$ = 10.2/s, $\boldsymbol{\mu_d}$ = 90 ms, $\boldsymbol{k}$ = 8.7, $\boldsymbol{\delta}$ = 0.60, $\boldsymbol{\sigma}$ = 0.006.

Decoding read-out, On/Off labelling and statistics

The decoder z score was modelled as $z(t) = c\,\Delta r(t) + \eta(t)$, where Δr is the cued minus uncued rate in 100 ms windows stepped by 10 ms. The noise η has unit variance, the autocorrelation of overlapping windows, and a slow per-trial offset whose variance share was measured from the pre-cue z of the data (0.35 for all neurons, 0.09 for the most selective neurons, 0.11 for random neurons). The gain c was not fitted: for each decoded population, it was chosen so that the upper mean of a two-component Gaussian mixture of delay z values matched the data's. We call this upper mean the decoding strength. States were labelled as in Panichello et al. (2024). On states are runs of $z > 1.96$ whose summed z exceeds the 95th percentile of the same quantity in pure noise. Off states are runs of at least 5 points (50 ms) with $z < 0.84$. States were counted from 0 to 1,400 ms after cue onset, and fractions of the delay spent On or Off from 500 to 1,400 ms. Model statistics were computed from 2,112 simulated trials (792 per evaluation during fitting). Data statistics are means over the 25 sessions; for random neurons, the 10 draws were first averaged within each session.

References

Daniel J. Amit and Nicolas Brunel. Model of global spontaneous activity and local structured activity during delay periods in the cerebral cortex. *Cerebral Cortex*, 7:237–252, 1997.

Edward Awh and John Jonides. Overlapping mechanisms of attention and spatial working memory. *Trends in Cognitive Sciences*, 5:119–126, 2001.

Joao Barbosa et al. Interplay between persistent activity and activity-silent dynamics in the prefrontal cortex underlies serial biases in working memory. *Nature Neuroscience*, 23:1016–1024, 2020.

Niko A. Busch and Rufin VanRullen. Spontaneous EEG oscillations reveal periodic sampling of visual attention. *Proceedings of the National Academy of Sciences USA*, 107:16048–16053, 2010.

Niko A. Busch, Julien Dubois, and Rufin VanRullen. The phase of ongoing EEG oscillations predicts visual perception. *Journal of Neuroscience*, 29:7869–7876, 2009.

Albert Compte, Nicolas Brunel, Patricia S. Goldman-Rakic, and Xiao-Jing Wang. Synaptic mechanisms and network dynamics underlying spatial working memory in a cortical network model. *Cerebral Cortex*, 10:910–923, 2000.

Ian C. Fiebelkorn and Sabine Kastner. A rhythmic theory of attention. *Trends in Cognitive Sciences*, 23:87–101, 2019.

Ian C. Fiebelkorn, Mark A. Pinsk, and Sabine Kastner. A dynamic interplay within the frontoparietal network underlies rhythmic spatial attention. *Neuron*, 99:842–853, 2018.

Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9:159–195, 2001.

Randolph F. Helfrich et al. Neural mechanisms of sustained attention are rhythmic. *Neuron*, 99: 854–865, 2018.

Ayelet N. Landau and Pascal Fries. Attention samples stimuli rhythmically. *Current Biology*, 22: 1000–1004, 2012.

Yuanyuan Mi, Mikhail Katkov, and Misha Tsodyks. Synaptic correlates of working memory capacity. *Neuron*, 93:323–330, 2017.

Gianluigi Mongillo, Omri Barak, and Misha Tsodyks. Synaptic theory of working memory. *Science*, 319:1543–1546, 2008.

Rana Mozumder, Zhengyang Wang, Wenhao Dang, Junda Zhu, Benjamin M. Hammond, Anna Machado, and Christos Constantinidis. Asynchronous firing and off states in working memory maintenance. *Cell Reports*, 45:116764, 2026.

Matthew F. Panichello and Timothy J. Buschman. Shared mechanisms underlie the control of working memory and attention. *Nature*, 592:601–605, 2021.

Matthew F. Panichello et al. Intermittent rate coding and cue-specific ensembles support working memory. *Nature*, 636:422–429, 2024.

Mark G. Stokes. 'activity-silent' working memory in prefrontal cortex: a dynamic coding framework. *Trends in Cognitive Sciences*, 19:394–405, 2015.

Weicheng Tao and Camilo Libedinsky. Evidence of activity-silent working memory in prefrontal cortex. *bioRxiv*, 2024. doi:10.1101/2024.06.03.597259.

Stefano Vicini et al. Functional and pharmacological differences between recombinant N-methyl-D-aspartate receptors. *Journal of Neurophysiology*, 79:555–566, 1998.

Min Wang et al. NMDA receptors subserve persistent neuronal firing during working memory in dorsolateral prefrontal cortex. *Neuron*, 77:736–749, 2013.

Xiao-Jing Wang. Synaptic basis of cortical persistent activity: the importance of NMDA receptors to working memory. *Journal of Neuroscience*, 19:9587–9603, 1999.

Xiao-Jing Wang. Synaptic reverberation underlying mnemonic persistent activity. *Trends in Neurosciences*, 24:455–463, 2001.

Xiao-Jing Wang. Probabilistic decision making by slow reverberation in cortical circuits. *Neuron*, 36:955–968, 2002.

Kong-Fatt Wong and Xiao-Jing Wang. A recurrent network mechanism of time integration in perceptual decisions. *Journal of Neuroscience*, 26:1314–1328, 2006.